

Sender–Receiver Exercise 7: Reading for Receivers

Harvard SEAS - Spring 2026

2026-04-27

1 Sender–Receiver Exercise: IS LINEAR TIME is unsolvable

The goals of this exercise are:

- to develop your skills at understanding, distilling, and communicating proofs and the conceptual ideas in them,
- to practice reductions for proving unsolvability, and gain more intuition for what kinds of problems about programs are unsolvable.

To prepare for this exercise as a Receiver, you should try to understand the theorem statement and definition in Section 1.1 below, and review the material from the previous sections of this chapter. Your partner Sender will communicate the proof of Theorem 1.1.

1.1 The Result

Rice’s Theorem (Theorem 21.16 in the textbook) says that all nontrivial problems about the input–output behavior of programs (i.e. about a program’s semantics) are unsolvable. Here we will see an example of a computational problem that is *not* about the input–output behavior of programs but is nevertheless unsolvable:

Input: A RAM program P

Output: yes if P has running time $O(n)$, no otherwise

Computational Problem IS LINEAR TIME

As usual, the statement “ P has running time $O(n)$ ” means that there are constants c and n_0 such that for all $n \geq n_0$ and all inputs x of length at most n , $P(x)$ halts within $c \cdot n$ steps. Note that the constants c and n_0 are allowed to depend on P .

Theorem 1.1. IS LINEAR TIME *is unsolvable*.

The approach to proving unsolvability.

The reduction.

Correctness of the reduction.